# MD Jawadul Hasan

Turku, Finland | +358 415802333 | mdjawadulhasan@gmail.com
linkedin.com/in/mdjawadulhasan | github.com/mdjawadulhasan | mdjawadulhasan.github.io

## Profile

Software engineer with three years of experience building data driven enterprise systems, backend services, and AI applications. Passionate about developing scalable, intelligent systems. Currently exploring how to make AI agent outputs more trustworthy by tracing answers to source evidence and distinguishing computed results from model-generated inferences.

## Education

**Åbo Akademi University, Turku, Finland** August 2026 – Present

*M.S. in Computer Science, Erasmus Mundus Joint Master's Programme (EDISS)*

- Research topic (in collaboration with CNRS, France): Building a skepticism layer for LLM-based agentic systems that shows where each answer comes from, separates results computed from source data from model-generated inferences, and surfaces supporting evidence, so users can spot hallucinated claims and decide when to trust the system.
- Relevant coursework: Data Science, Cloud Computing, Artificial Intelligence, GPU Programming

**American International University - Bangladesh** September 2023

*BSc. in Computer Science and Engineering (CSE)* *Cumulative GPA: 3.99/4.00*

- Relevant coursework: Algorithms & Data Structures, OOP, Database, Networks, Web Technologies, Data Mining, Advanced Programming with Java & .NET, Software Quality & Testing, Artificial Intelligence

## Professional Experience

**Software Engineer** March 2023 – July 2026

*Brain Station 23 PLC, Dhaka, Bangladesh*

- Lead a team of **6** engineers delivering software solutions for large-scale enterprise clients across government services and international insurance sectors in highly regulated domains.
- Built an AI Agent integrating LLMs APIs to automate insurance claim processing, achieving **40%** faster submission times.
- Developed a conversational AI assistant that personalizes customer onboarding by inferring preferences from natural language, reducing manual intake effort significantly.
- Automated approval workflows supporting **10,000+** daily approvals, boosting operational efficiency through concurrency control and duplicate prevention.
- Built an integrated wallet and bank API system processing over **$3 million** in monthly payments and streamlining automated fee collection.
- Migrated **15** years of relational data with full schema mapping to improve report accuracy and reliability by **70%.**
- Designed and optimized data reporting solutions over tables of more than **50 million** records each.
- Architected a polyglot persistence strategy using MongoDB and SQL Server, reducing query latency from **12s to 500ms.**
- Migrated a legacy application to the latest .NET with comprehensive unit tests, reducing bugs by **30%**.
- Automated CI/CD pipelines with Jenkins and integrated OAuth authentication for secure third-party API connectivity.

**Research Assistant (Part-time)** December 2024 – Present

*ELITE Research Lab, NY, USA (Remote)*

- Pursuing applied research in machine learning and AI, focusing on generative models, predictive modeling, and computer vision.
- Developing AI-driven systems and tools, including context-aware conversational agents for real-world applications.
- Mentoring undergraduate students in LLM, AI, and machine learning research, and supporting the development of research tools and AI applications.

## Skills

**Core Competencies**: Problem Solving, Algorithms & Data Structures, OOP, Agile, Scrum

**Programming**: C#, C++, Python, SQL, Java, JavaScript

**Frameworks & Libraries**: .NET, xUnit, Angular, Nest Js, Next Js

**AI & ML**: RAG, Vector Search, Embedding Models, TensorFlow, PyTorch

**Database**: MSSQL, MongoDB, PostgreSQL

**DevOps & Cloud:** Git, CI/CD, Jenkins, Docker, Azure, AWS

**Architecture & Design**: Clean Architecture, Microservice, Vertical Slice, MVC, REST, CQRS, SOLID, DRY, YAGNI

## Projects

**GradAgent.ai** | *AI-Powered Graduate Admissions Assistant*

- Built an ASP.NET Core backend for an admissions chatbot, grounding LLM answers in MongoDB vector search over university and professor data.
- Implemented dynamic prompt construction and context injection to optimize LLM responses based on retrieved user profiles, university data, and application requirements..
- Developed an intelligent, tool-calling LLM agent to generate personalized university application strategies based on individual student profiles and requirements..
- Integrated RAG pipelines over a curated knowledge base of university requirements, admission statistics, and program details to ground agent responses in accurate, up-to-date information.
- Engineered a sliding window mechanism for token allocation and rate limiting to control LLM costs across API .

## Publications

- **Md Jawadul Hasan**, S. H. Shifat, J. Matubber, R. Hossain, M. A. Rahman, B. M. T. Haque, M. J. Hossen. "An in-depth exploration of machine learning methods for mental health state detection: a systematic review and analysis." *Frontiers in Digital Health*, vol. 7, 2026.
- **Md Jawadul Hasan**, A. Das, J. Matubber, S. H. Shifat, M. K. Morol. "Enhanced Classification of Anxiety, Depression, and Stress Levels: A Comparative Analysis of DASS21 Questionnaire Data Augmentation and Classification Algorithms." *3rd International Conference on Computing Advancements (ICCA 2024)*, ACM.
- M. A. H. Reaj, R. D. Gupta, M. Y. Rahat, N. Fahad, **Md Jawadul Hasan**, T. H. Liew. "GANDiff-FR: Hybrid GAN–Diffusion Synthesis for Causal Bias Attribution in Face Recognition."

## Certifications

- Google Data Analytics Professional Certificate – Google (Coursera)

## Achievements

- Ranked 6th in the Intra AIUB Programming Contest Fall 2021–22
- Solved 600+ algorithmic challenges on LeetCode, Codeforces, HackerRank, and similar platforms
- Selected for the Dean's List honors for outstanding academic performance
- Finalist in an app development competition hosted by IEEE India in 2020
- Top 1% in class; received Summa Cum Laude for an exceptional CGPA

## Languages

**English**: Fluent

**Finnish**: Learning, currently enrolled in Finnish for Foreigners 1, Metropolia University of Applied Sciences